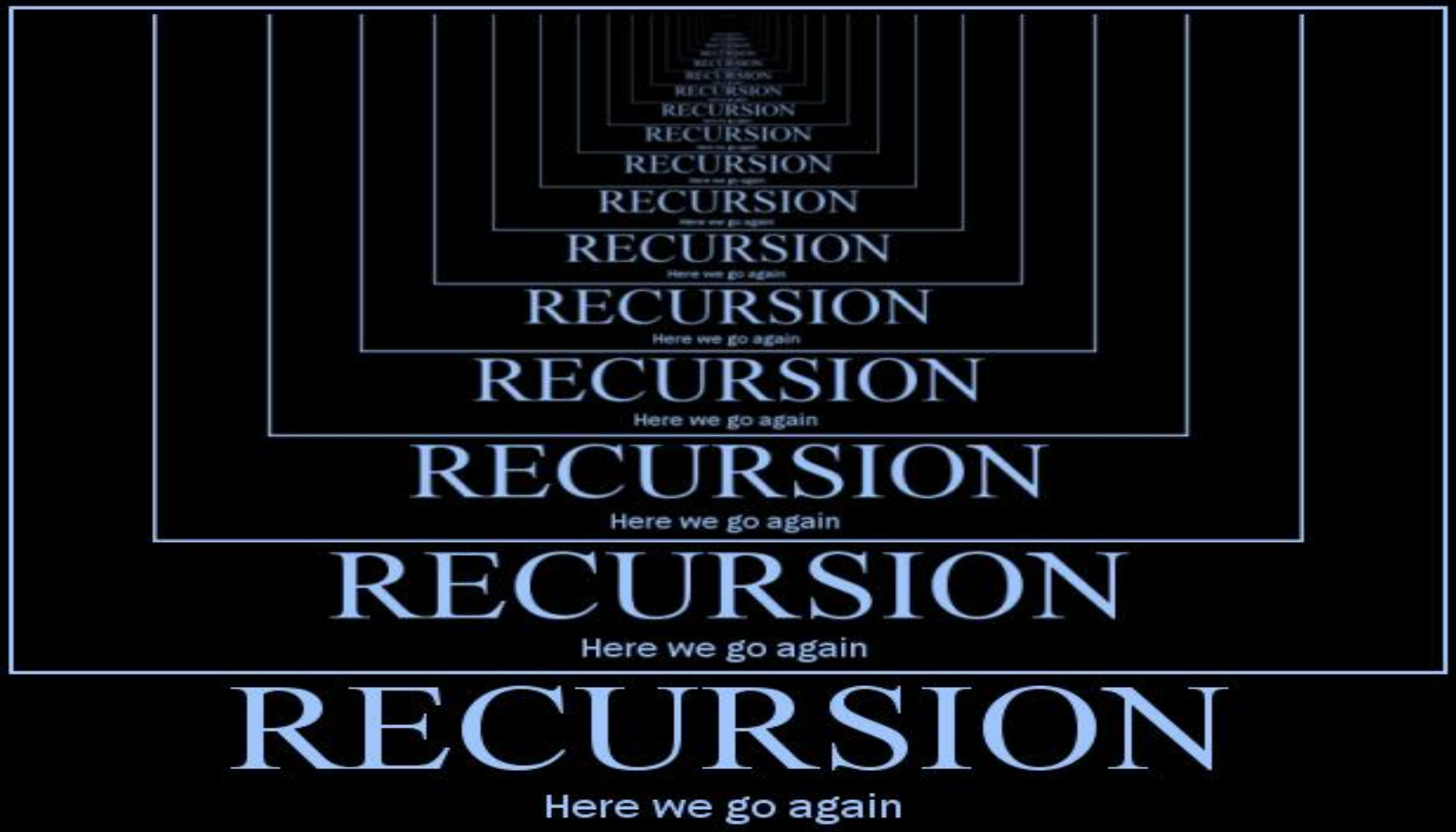


# Java Programming

CPT111 – Week 11

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# CPT111 Java Programming

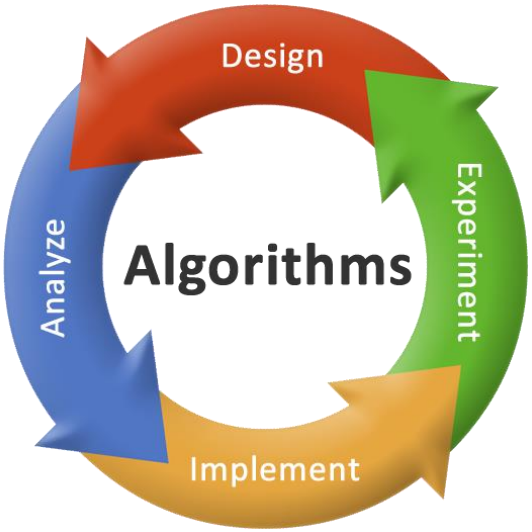
## Week 11

# Recursion

# Learning Outcomes

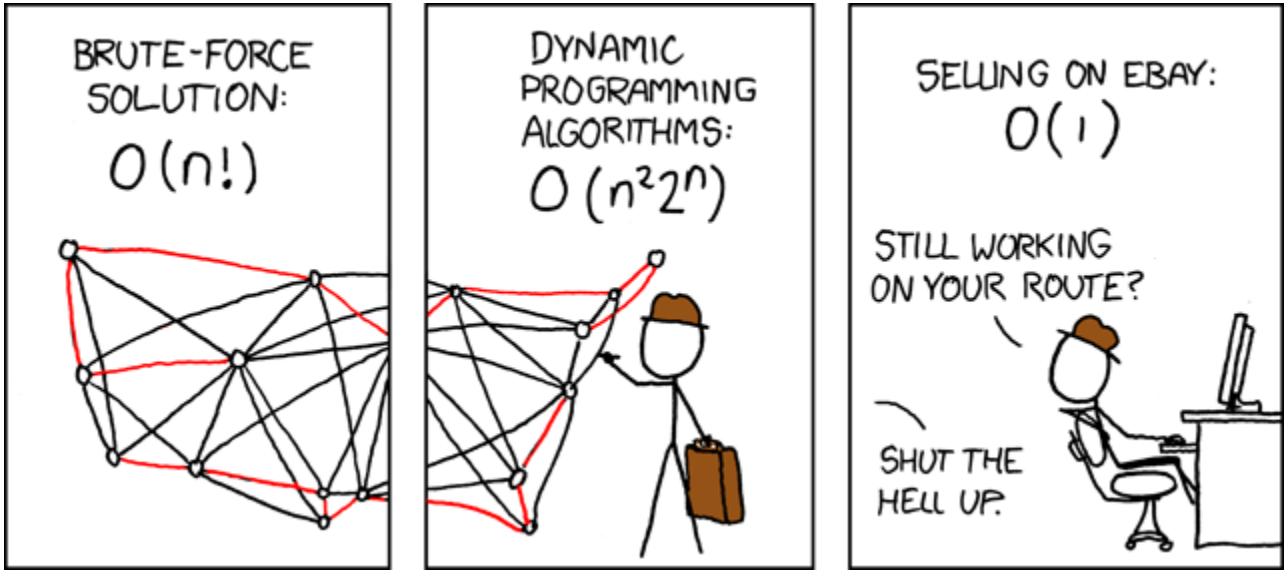
- Recursion
  - Factorial
  - Recursion God  vs  Recursive Calls
  - Handshake Problem
- Recursion Recipe
  - Base Case
  - Recursive Step
  - with Helper Method
- More examples
  - Fibonacci Number,  The Twelve Days of Christmas, Subsequences
- Representing Date and Time

# Algorithms Design Techniques




- Recursion is one of the tools in designing algorithm
- Transition from coding to problem solving
- Designing **fast** algorithm
  - Divide-and-Conquer := Divide + Delegate (Recursion) + Combine
  - Dynamic Programming := Recursion + Memoization
  - will learn in *Algorithms and Data Structure courses*

# Travelling Salesman Problem

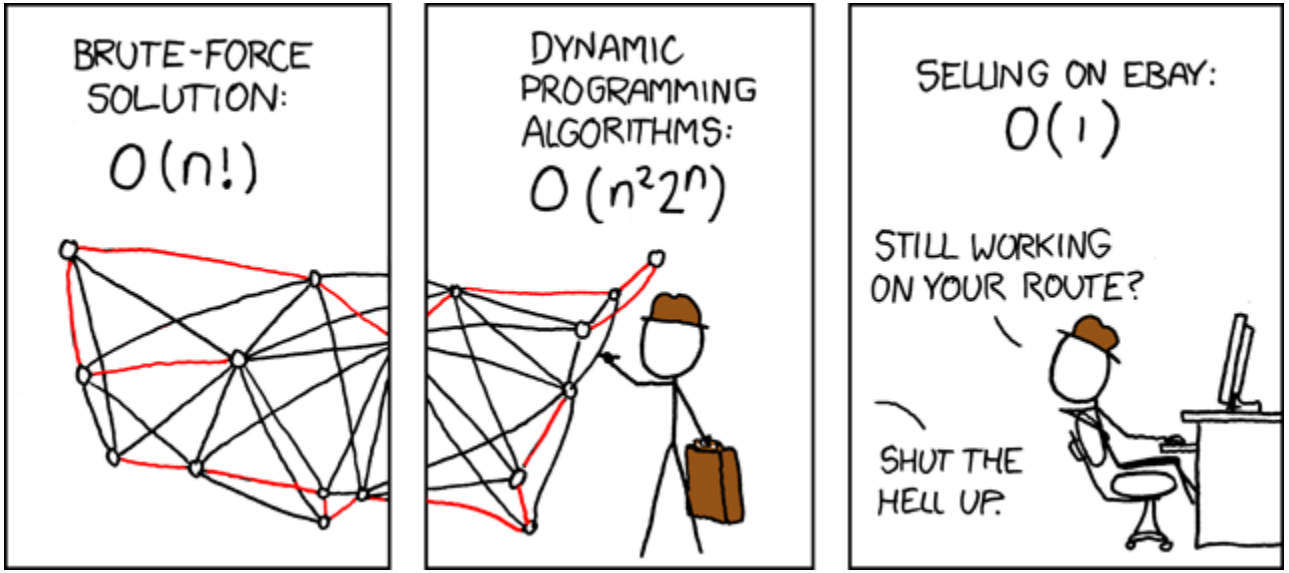



- Asks what is the shortest tour that visits *all* n cities
- Brute-force solution: try all tours
- How many possible tours are there ? n ! possible tours

旅行商问题

# Factorial

暴力解法 动态规划




- Factorial `fact(n)` of a number n >= `0` is defined as follows
  - `fact(0) = 1`
  - `fact(n) = n * n-1 * n-2 * ··· * 3 * 2 * 1`

# In-Class Quiz 1: Factorial Number (1)

What is factorial number of 4, `fact(4)` ?

a) `fact(4) = 4`

b) `fact(4) = 12`

c) `fact(4) = 24`

# In-Class Quiz 2: Factorial Number (2)

What is factorial number of 5, `fact(5)` ?

a) `fact(5) = 5`

b) `fact(5) = 24`

c) `fact(5) = 120`

# In-Class Quiz 3: Factorial Number (3)

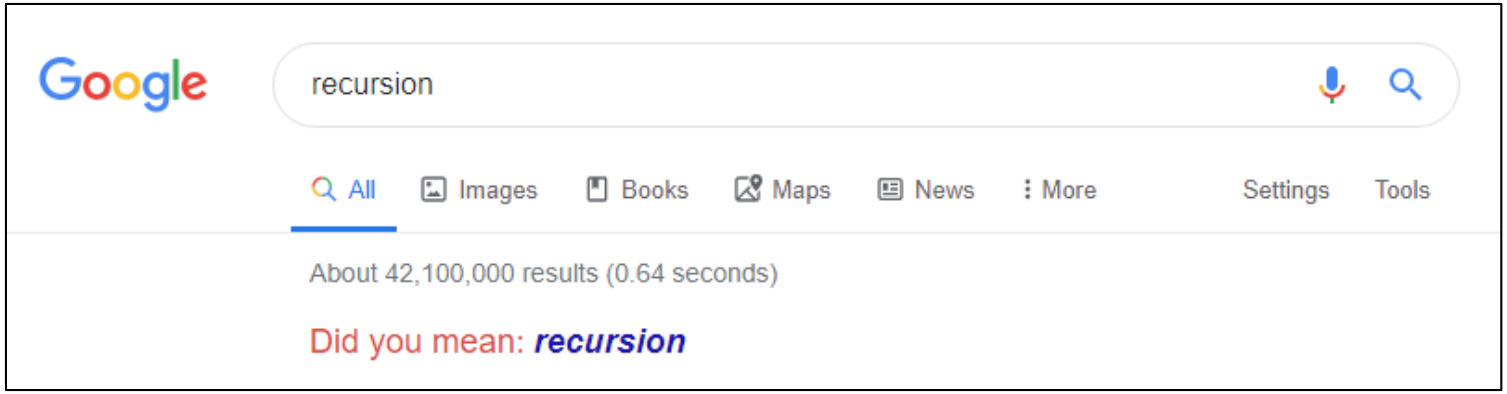
What is factorial number of n, `fact(n)` ?

a) `fact(n) = n * (n-1)`

b) `fact(n) = fact(n-1) * n`

c) `fact(n) = fact(n-1) * (n-1)`

# Recursive Factorial




- Factorial `fact(n)` of a number `n >= 0` is *recursively* defined as follows
  - `fact(0) = 1`
  - `fact(n) = fact(n-1) * n`

# Recursion Code

```
public static long fact(int n) {
    if
        return
    else
        return
}
```

```
public static void main(String[] args) {


}
```

# Recursion God ?!

```
public static long fact(int n) {
    if (n == 0)
        return 1;
    else
        return n * fact(n-1);
}
```

Recursion God

- *Recursion God* will tell you the answer of `fact(n-1)` !!
  - Magic ?!

# Inception

Have you guys watched Inception ?

Warning ! Major SPOILER ahead !

# Inception World : Sharing Dream inside a Dream ! (1)

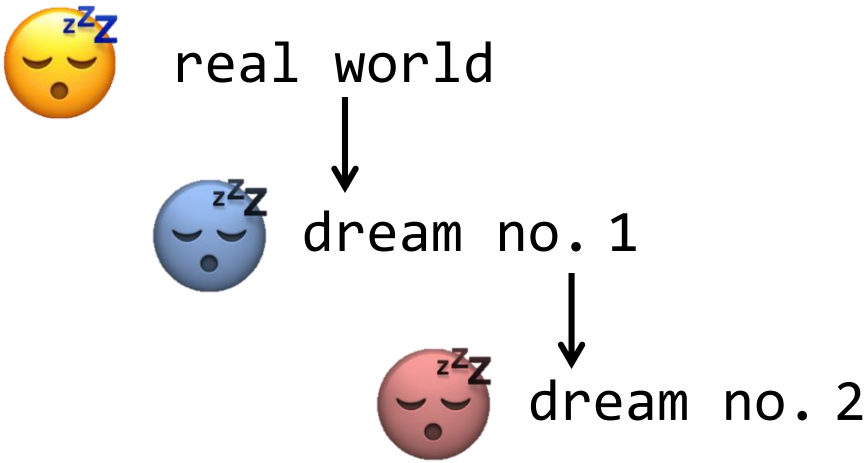
real world

↓

dream no. 1

Dream Machine

# Inception World : Sharing Dream inside a Dream ! (1)



Dream Machine

# Inception World : Sharing Dream inside a Dream ! (1)

real world

↓

dream no. 1

↓

dream no. 2

↓

dream no. 3

↓

dream no. 4

↓

dream no. 5

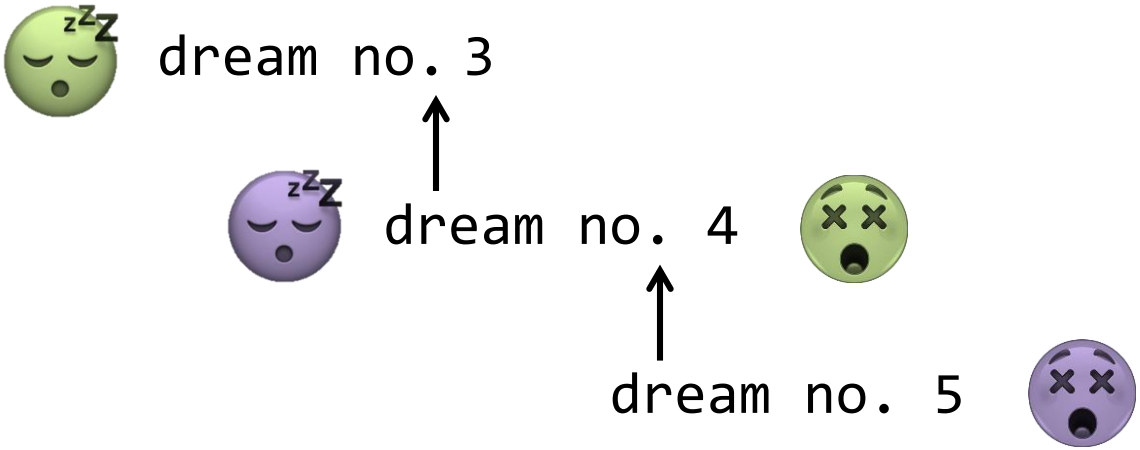
Dream Machine

# Inception World : Sharing Dream inside a Dream ! (2)

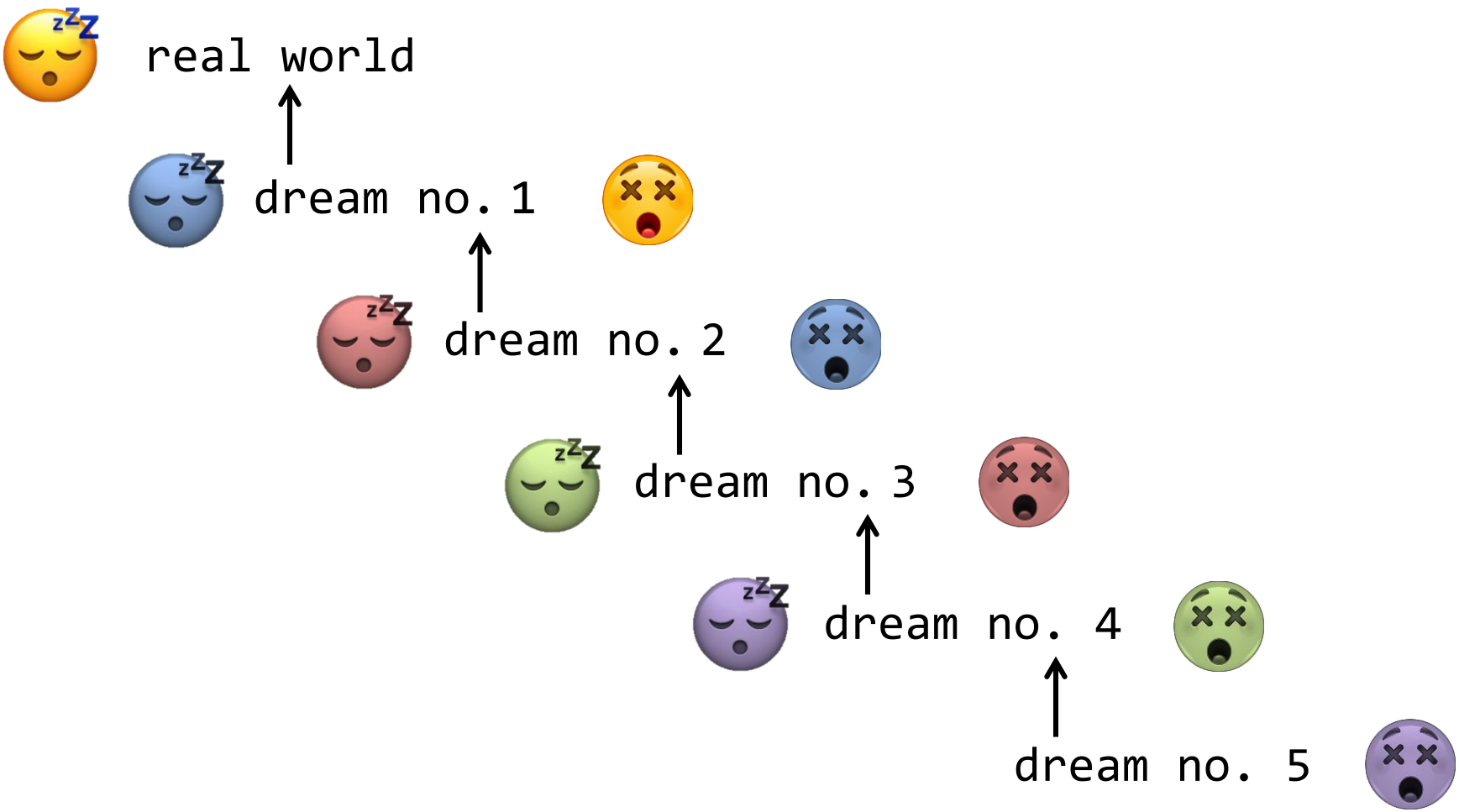
dream no. 4

dream no. 5

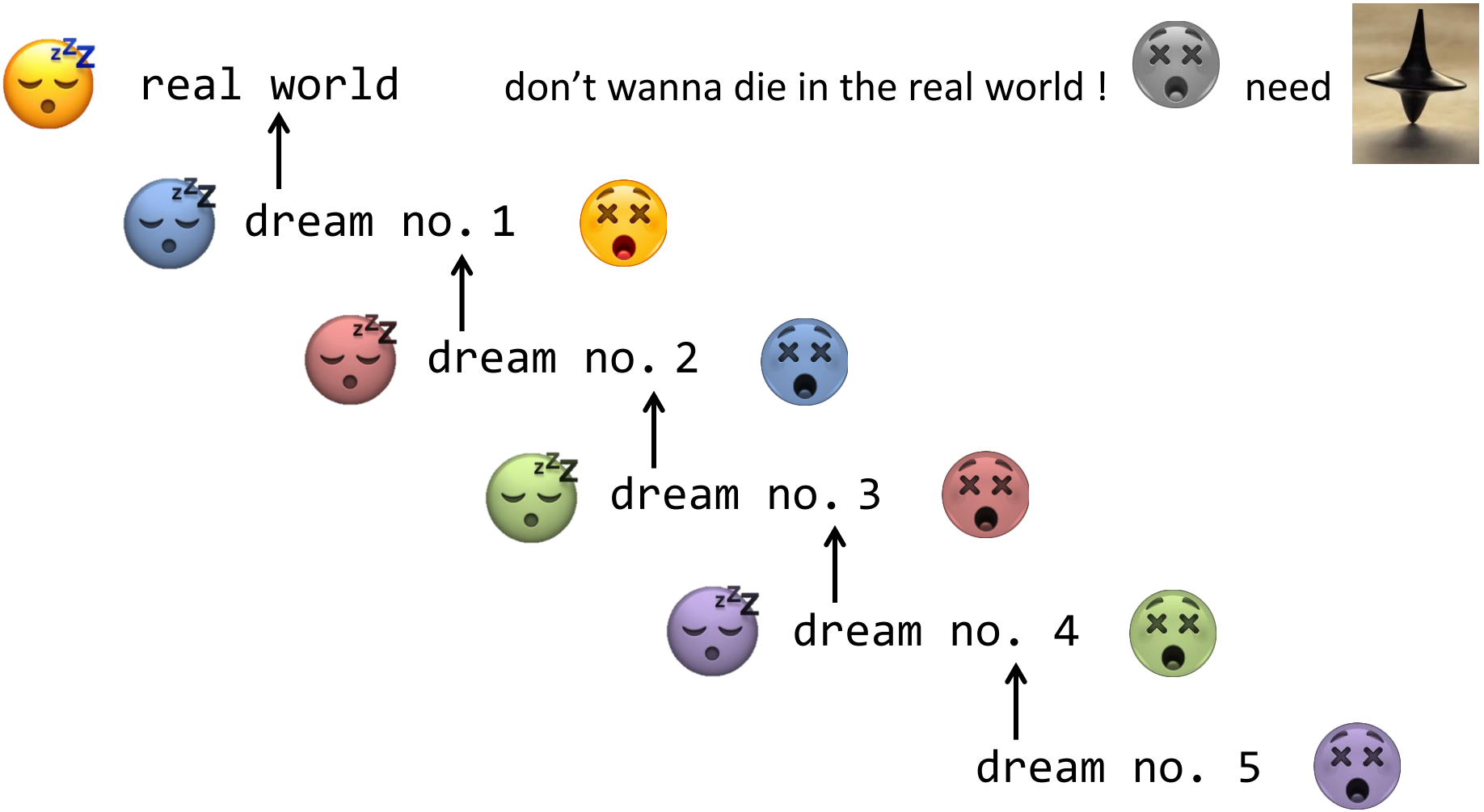
# Inception World : Sharing Dream inside a Dream ! (2)



# Inception World : Sharing Dream inside a Dream ! (2)



# Inception World : Sharing Dream inside a Dream ! (2)



# Factorial Recursive Calls (1)

```
fact(5)
  ↓
return 5 * fact(4)
```

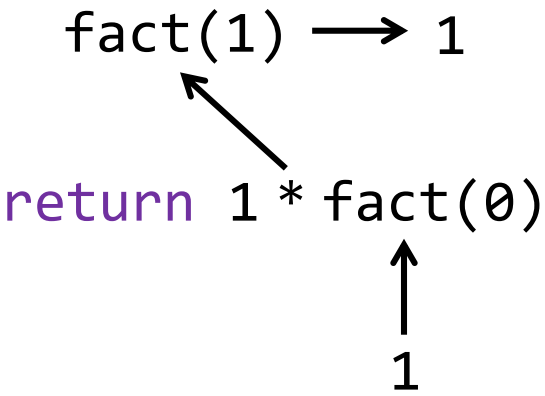
# Factorial Recursive Calls (1)

```
fact(5)
  ↓
return 5 * fact(4)
              ↓
         return 4 * fact(3)
```

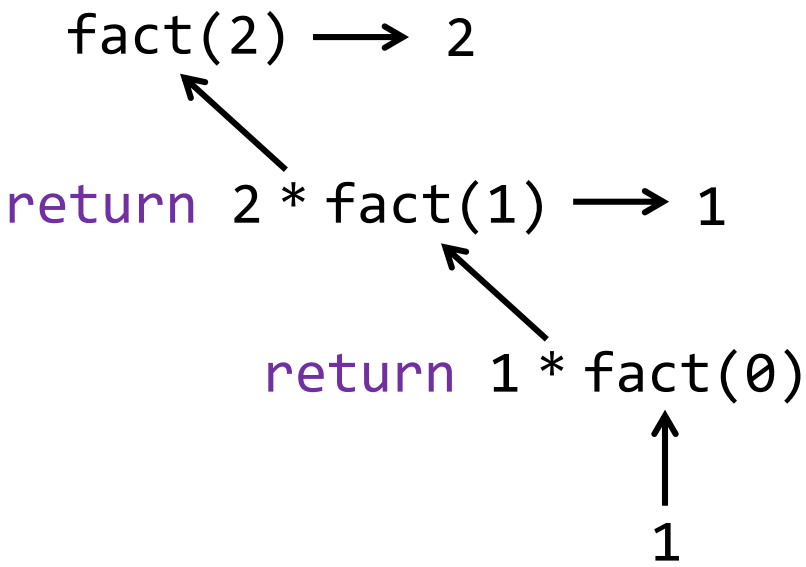
# Factorial Recursive Calls (1)

```
fact(5)
  ↓
return 5 * fact(4)
            ↓
        return 4 * fact(3)
                    ↓
                return 3 * fact(2)
                            ↓
                        return 2 * fact(1)
                                    ↓
                                return 1 * fact(0)
                                            ↓
                                            1
```

# Factorial Recursive Calls (2)



# Factorial Recursive Calls (2)



# Factorial Recursive Calls (2)

```
fact(5) ⟶ 120

return 5 * fact(4) ⟶ 24

        return 4 * fact(3) ⟶ 6

                return 3 * fact(2) ⟶ 2

                        return 2 * fact(1) ⟶ 1

                                return 1 * fact(0)

                                        1
```

# The Handshake Problem

- There are n people in a room
- If each person shake hands *once* with every other person, what is the total number of handshakes `handshake(n)` ?

# In-Class Quiz 4: Handshake Problem : Base Case

What is the base case of the Handshake Problem ?

a) `handshake(2) = 0`

b) `handshake(2) = 1`

c) `handshake(2) = 2`

# In-Class Quiz 5: Handshake Problem : Recursive Step

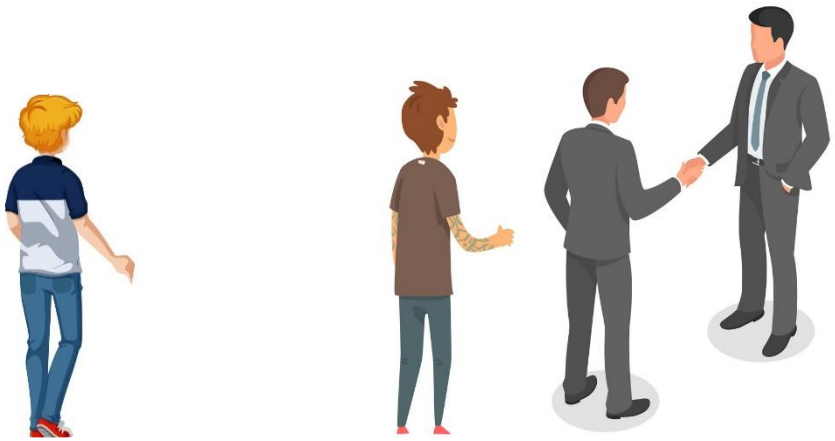
A third person comes. What is the number of handshakes now ?

a) `handshake(3) = handshake(2) + 3`

b) `handshake(3) = handshake(2) + 2`

c) `handshake(3) = handshake(2) + 1`

# The Handshake Problem : Towards Recursive Step



A fourth person comes.  What is the number of handshakes now ?

`handshake(4) =`

# Complete The Handshake Problem Code !

```
public static int handshake(int n) {



}
```

# Recipe for Recursion

The problem can be **reduced** into smaller and easier subproblems

<=> the solution to a problem **depends** on the solutions to smaller instances of the same problem

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1. Base Case :
   - Find the simplest instance that can be **directly** solved

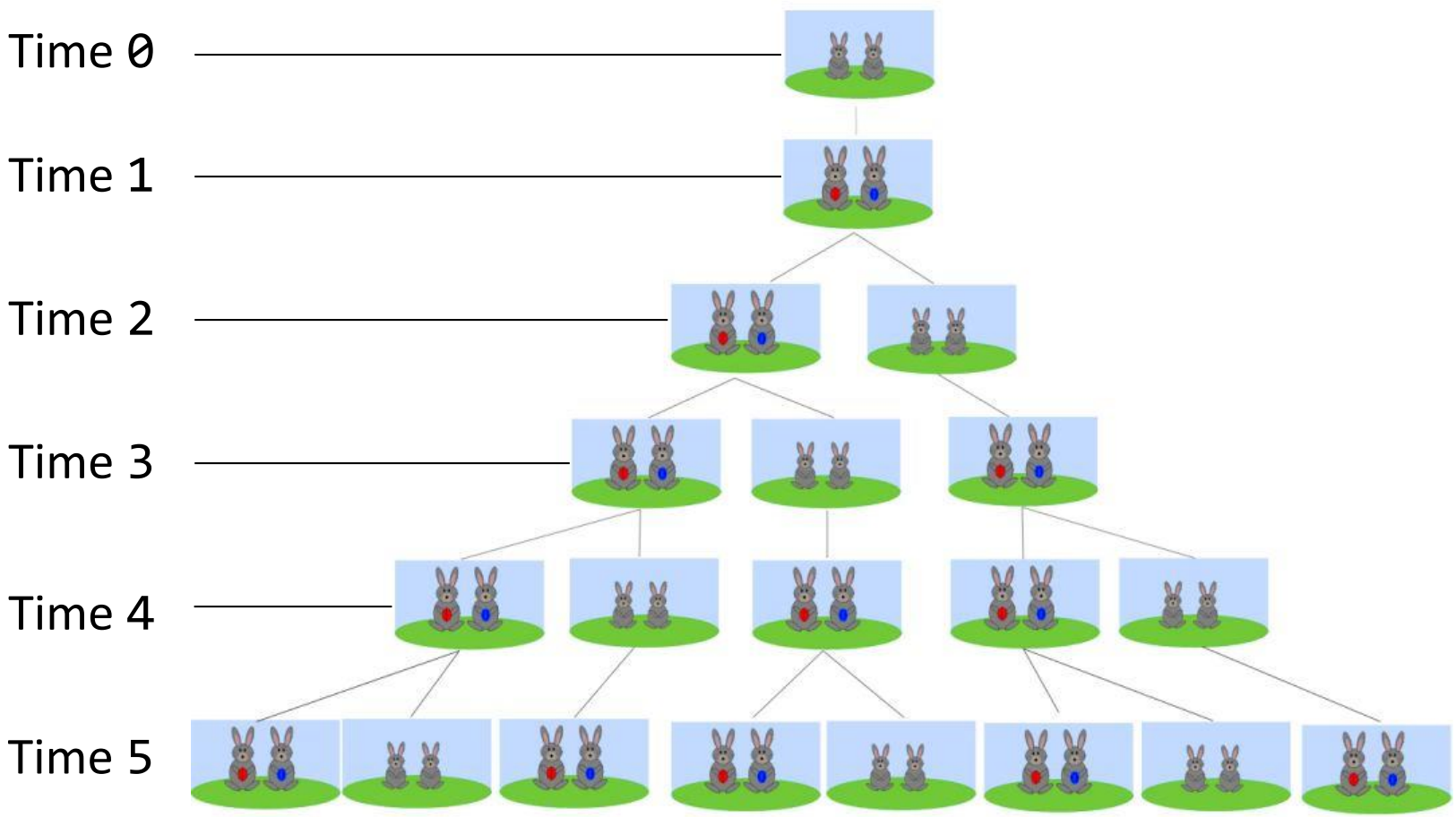
# Recipe for Recursion

The problem can be **reduced** into smaller and easier subproblems

<=> the solution to a problem **depends** on the solutions to smaller instances of the same problem

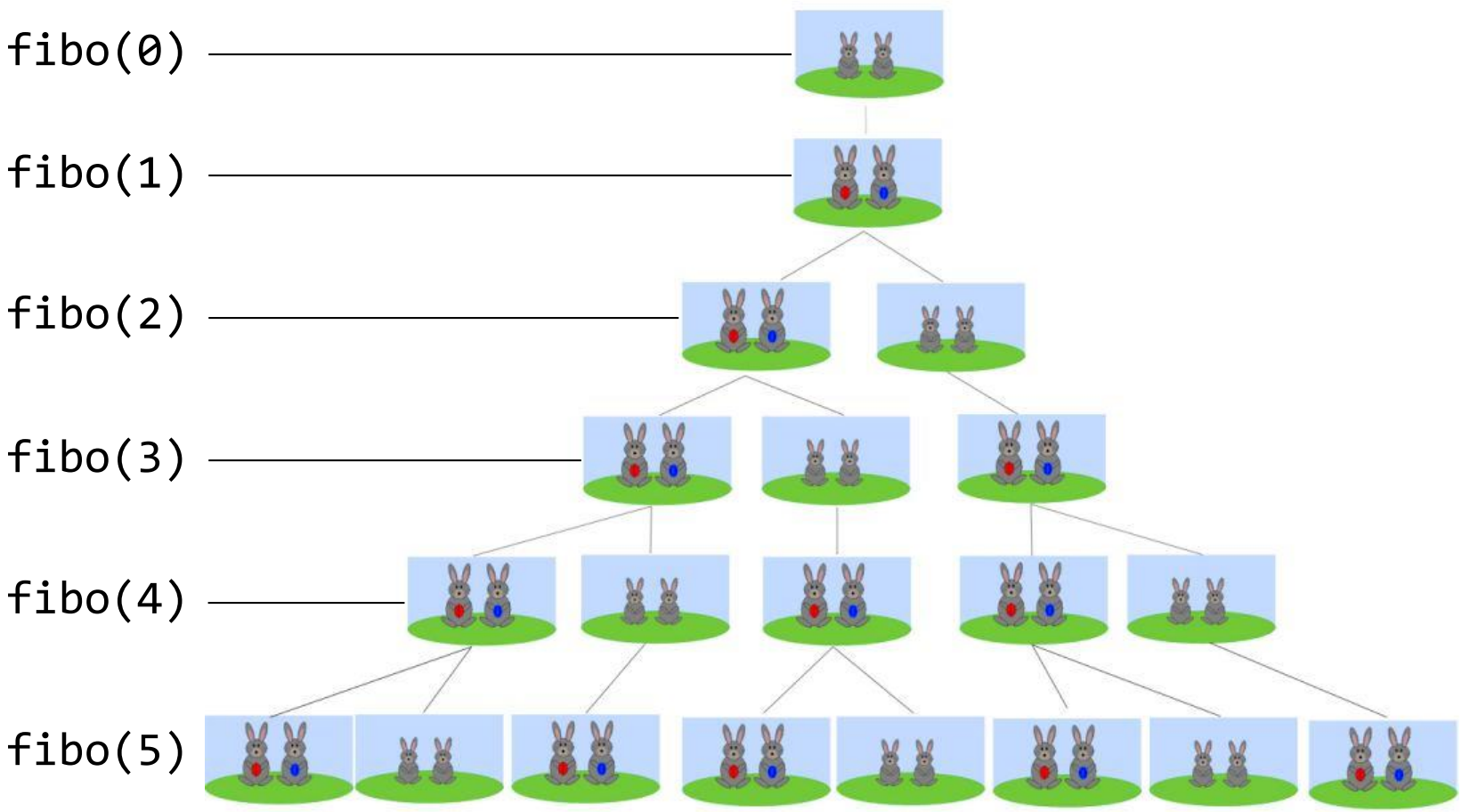
1. Base Case :
   - Find the simplest instance that can be **directly** solved
2. Recursive Step :
   - Try a couple of cases to see the recursive **relation** among instances
   - Formulate generally for n,
     trust Recursion God to solve the smaller instances
   - Make sure to reach the base case

# Rabbit Breeding Problem



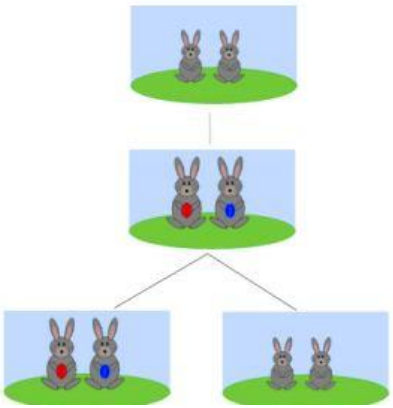

- Assume a pair of rabbits take one time period to mature and breed, always conceive a male female pair, never die
- How many pairs of rabbits after time period Time n ?

# Fibonacci Number




- What is the Fibonacci number `fibo(n)` ?

# In-Class Quiz 6: Fibonacci Number : Base Case



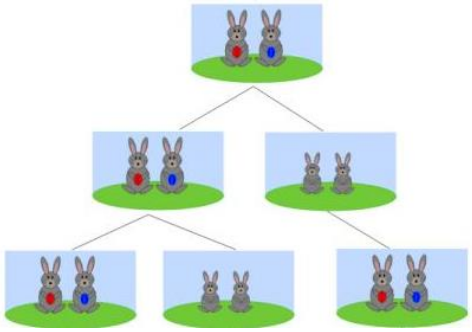
What is the base case of the Fibonacci Number ?

a) `fibo(0) = 1`

b) `fibo(1) = 1`

c) `fibo(0) = 1 and fibo(1) = 1`

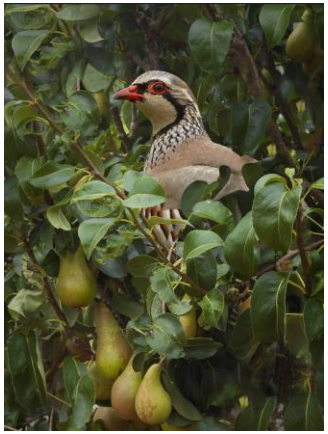
# Complete the Fibonacci Number Code!



```
public static int fibo(int n) {

}
```

# The Twelve Days of Christmas



- On the first day of Christmas
  My true love gave to me
  A partridge in a pear tree.

- On the second day of Christmas
  My true love gave to me
  Two turtle doves,
  And a partridge in a pear tree.

- On the third day of Christmas
  My true love gave to me
  Three French hens,
  Two turtle doves,
  And a partridge in a pear tree.

# The Twelve Days of Christmas

- On the fourth day of Christmas
  My true love gave to me
  Four calling birds,
  Three French hens,
  Two turtle doves
  And a partridge in a pear tree.

- On the fifth day of Christmas
  My true love gave to me
  Five golden rings,
  Four calling birds,
  Three French hens,
  Two turtle doves
  And a partridge in a pear tree ...

- Use recursion to compute ***how many*** gifts your true love gave to you on the ***n***th day of Christmas!
  And, ***total*** gifts up to *n*th days of Christmas!

# The Twelve Days of Christmas

- Test cases:

| n | #gifts on day n | #total gifts on day n |
|---|---|---|
| 1 | 1 | 1 |
| 2 | 3 | 4 |
| 3 | 6 | 10 |
| 4 | 10 | 20 |
| 5 | 15 | 35 |
| 12 | 78 | 364 |
| 400 | 80200 | 10746800 |

- Complete the code for the Twelve Days of Christmas!

# Subsequences Problem

- Given a word consisting only of letters A-Z or a-z
- Return all subsequences of word, separated by commas, where a subsequence is a string of letters found in word in the same order that they appear in word.

```
subsequences("abc") -> "abc,ab,bc,ac,a,b,c,"
```

- Note the trailing comma preceding the empty string `""`, which is also a valid subsequence

# Recursion with Helper Methods

- For the Subsequences Problem, instead of solving it with *direct* recursive implementation,

```
public static String subseq(String word)
```

it is easier to use additional *Helper Method*

```
private static String subseqHelper(String
                                   partialSubseq, String word)
```

- Helper method introduces another parameter(s) to give you more flexibility in doing (more creative) recursion calls

# Subsequences with Helper Methods (1)

```
private static String subseqHelper(String
                                   partialSubseq, String word)
```

- Idea: we build up a partial subsequence using the initial letters of the word
  - then use recursive calls to complete that partial subsequence using the remaining letters of the word
- For example, word = "abc"
  - First, select "a" to be in the partial subsequence, and recursively extend it with all subsequences of "bc"
  - Then, use "" as the partial subsequence, and again recursively extend it with all subsequences of "bc"

# Subsequences with Helper Methods (2)

```
subsequences("abc") -> "abc,ab,bc,ac,a,b,c,"
```

- In other words, recursively build up subsequences that, either use the first letter `a` or not…

# Subsequences Code

```
private static String subseqHelper(String
                                   partialSubseq, String word) {
   if (word.isEmpty())
      // base case
      return partialSubseq;
   else
      // recursive step
      return subseqHelper(partialSubseq + word.charAt(0),
                                          word.substring(1))
            + ","
            + subseqHelper(partialSubseq,
                                          word.substring(1));
}
```

# Notes for Recursive + Helper Methods

- Don’t forget to implement the original problem

```
public static String subseq(String word)
    return subseqHelper("", word);
}
```

- You can use the same method name for your helper methods

- Hide the helper methods,  don’t expose it to the client

# Representing Date (not!)

- `java.util.Date`
  the Date class represents a date and a time as the number of milliseconds since epoch: midnight on January 1, 1970
- `java.util.Calendar`
  a Calendar is used to get information (e.g. the day of the week) about a Date object

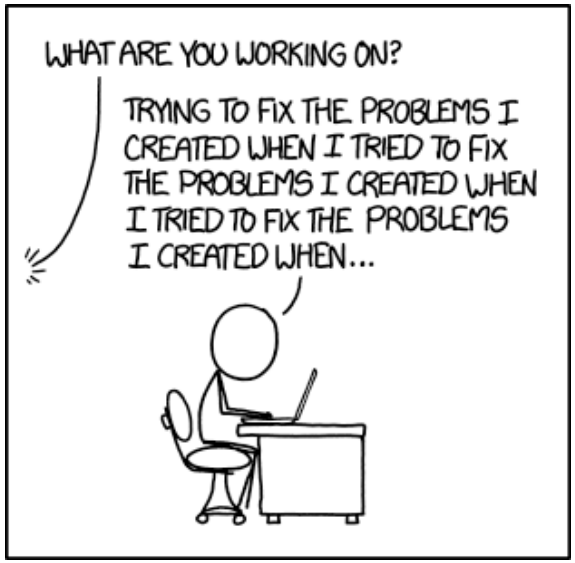
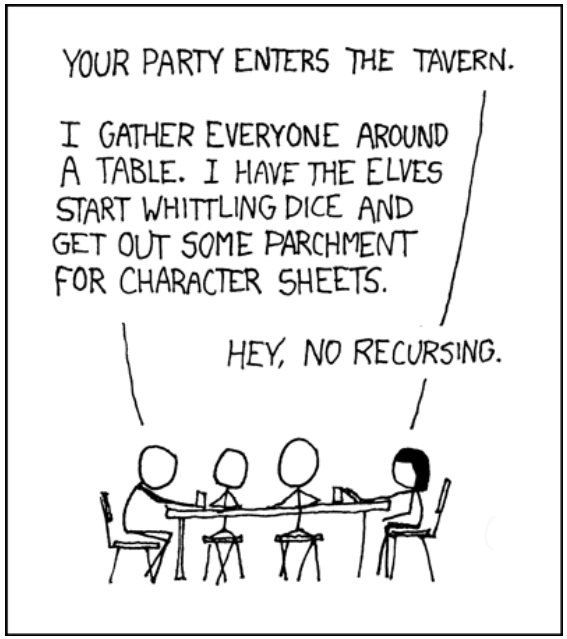
- They should ***no*** longer be used:
  - Confusing and badly designed API, for example:
    Years are "year − 1900" (so 2025 is 125)
    Months are 0-based (January = 0, December = 11)
  - Both are mutable
    Any method that pass them to can silently change the value
  - Deprecated

# Representing Date

- `java.time` APIs address deficiencies of `Date` and `Calendar`
  - immutable, support leap years/seconds, nanosecond-granularity
- Popular packages:
  - `java.time`: Dates, times, time zones
  - `java.time.zone`: Rules for time zone calculations
  - `java.time.temporal`: APIs for modifying dates and times
  - `java.time.format`: Formatting dates and times
- Fundamental date/time classes in `java.time`:
  - `Instant` is a point in time (number of milliseconds since the epoch)
  - `LocalDate`, `LocalTime`, and `LocalDateTime` represent a date and/or time without a timezone
  - `ZonedDateTime` is a date/time with a specific time zone and all of its associated rules
  - `ZoneId` represents a time zone
  - Portions of a date are represented with the `Year`, `Month`, `MonthDay`, etc. classes
  - `Duration` measures an amount of time in days/hours/minutes etc.

see https://www.baeldung.com/java-8-date-time-intro

# Tasks of Week 11

- Lab 11
- Exercise and CW1 Week 11
- Watch Inception (again) !

Thank you for your attention !